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The Kyshtym disaster: a catastrophe disguised as the northern lights

60 years since the Kyshtym accident

Alla Salkova



A sign in the village of Muslyumovo, damaged by the Mayak chemical plant accident in 1957, 2010

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Sixty years ago, the Kyshtym accident occurred, second only to the Chernobyl Nuclear Power Plant explosion and the Fukushima-1 tragedy in terms of the scale of radiation contamination. Gazeta.Ru examines the cause of the disaster and how the authorities concealed it from the public.

On September 29, 1957, at 4:00 PM, the first radiation accident in the USSR occurred at the Mayak Chemical Plant, located in the closed city of Chelyabinsk-40 (now Ozersk): a radioactive waste storage tank exploded. The disaster was named the Kyshtym accident, after the city of Kyshtym, the closest to Chelyabinsk-40.

The explosion occurred in a 300 cubic meter tank due to a failure of the cooling system. The tank contained a total of approximately 80 cubic meters of highly radioactive nuclear waste. When it was built in the 1950s, the structure's structural integrity was beyond doubt. It was located in a pit, enclosed in a meter-thick concrete jacket.

The tank's lid weighed 560 tons, and a two-meter-thick layer of earth was placed on top of it. However, even this couldn't contain the explosion.

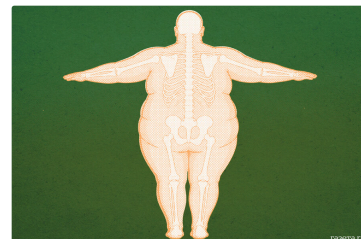
According to another, unofficial version, the disaster occurred due to an error by plant employees who mistakenly added plutonium oxalate solution to a evaporation tank containing hot plutonium nitrate solution. When the nitrate oxidized the oxalate, a large amount of energy was released, leading to overheating and explosion of the tank.

During the explosion, approximately 20 million Ci of radioactive substances

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Vera Kashtanova



kilometers and formed an aerosol cloud.

Over the next 11-12 hours, radioactive fallout fell over an area 300-350 km northeast of the explosion site.

The radiation contamination zone covered 23,000 square kilometers with a population of 270,000 in 217 settlements in the Chelyabinsk, Sverdlovsk, and Tyumen regions. During the cleanup, 23 villages with a population of 10,000-12,000 had to be resettled; all buildings, property, and livestock were destroyed.

Hundreds of thousands of military personnel and civilians became liquidators.

In the first ten days alone, the death toll from radiation reached hundreds; in total, 250,000 liquidators suffered to varying degrees during the work.

On the International Nuclear Test Scale, the accident was rated a level 6. By comparison, level 7, the maximum, was assigned to the Chernobyl and Fukushima Daiichi nuclear power plant accidents.

To prevent the spread of radiation, the government established a sanitary protection zone, where economic activity was prohibited. In 1968, the East Ural State Nature Reserve was established in this area.

Visiting it is prohibited - the level of radioactivity is still too dangerous for humans.

The reserve plays an important role in conducting scientific research on radiation.



Изображение временно недоступно

At the site of the explosion, a column of smoke and dust about a kilometer high formed, flickering with an orange-red light.

On October 6, 1957, the Chelyabinsk Worker newspaper published an article dedicated to him, which, however, did not say a word about the accident:

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"Last Sunday evening... many Chelyabinsk residents observed a special glow in

aurora borealis. An intense red glow, at times fading to faint pink and light blue, initially covered a significant portion of the southwestern and northeastern sky. Around 11 a.m., it could be observed in a northwesterly direction... Comparatively large colored areas and, at times, calm bands appeared against the sky, oriented meridionally in the final stages of the aurora. The study of the nature of the aurora borealis, begun by Lomonosov, continues to this day. Modern science has confirmed Lomonosov's fundamental idea that the aurora borealis occurs in the upper atmosphere as a result of electrical discharges... Aurora borealis... can be observed in the future at latitudes of the Southern Urals."

The Kyshtym disaster was a state secret for a long time. It was first openly discussed in films made in the late 1980s and early 1990s by director and biologist Elena [Sakanyan](#) , dedicated to the life of Soviet biologist and geneticist Nikolai Timofeev-Resovsky.

The films were shown on television only after Sakanyan directly requested that they be shown to [Boris Yeltsin](#) .

However, information leaked to the foreign press as early as April 1958. A Copenhagen newspaper first reported the accident. Subsequently, data on the accident appeared in a report from the US National Laboratory. Biologist Zhores Medvedev dedicated a book to the incident, titled "Nuclear Disaster in the Urals," and published it in the US. A group of American scientists from the Oak Ridge Nuclear Research Center also conducted an analysis of the accident and its causes.

For a long time, the public knew virtually nothing about the explosion at Mayak. Later, for some unknown reason, the accident was publicly reported in the media as the "Kyshtym accident."

An obelisk was even recently erected in Kyshtym to commemorate this event, although the city had nothing to do with it.

"And the East Ural radioactive trail that formed after 1957 did not affect Kyshtym and its residents," one of its liquidators said in an interview in 2009.

In total, more than 30 incidents involving radioactive emissions and human casualties were recorded at Mayak.

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